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EXPERIMENT NO – 8

Aim:

The aim of this project is to perform data preprocessing, apply classification, clustering, and association algorithms, and demonstrate their functionality using the Weka data mining tool. The goal is to explore how different data mining techniques can be applied to real-world datasets for extracting meaningful patterns, insights, and predictions.

Introduction:

Weka (Waikato Environment for Knowledge Analysis) is a powerful, open-source software suite that provides a collection of machine learning algorithms and tools for data mining tasks. It supports tasks such as classification, clustering, regression, and association rule mining, with an easy-to-use graphical interface for users. By leveraging Weka, we can preprocess data, apply machine learning models, and evaluate their performance efficiently.

Procedure:

1. Data Preprocessing:

Load Dataset: Import datasets in formats like `.csv` or `.arff` into Weka.

Clean Data: Handle missing values, remove irrelevant attributes, and normalize or standardize data.

Split Data: Divide the dataset into training and testing sets (e.g., 70-30 split or use cross-validation).

2. Classification:

Choose a classification algorithm (e.g., J48, Naive Bayes, Random Forest). Train the model on the training data and evaluate its accuracy on the testing data.

3. Clustering:

Select a clustering algorithm (e.g., K-means, EM).

Group the data into clusters based on similarity and evaluate the cluster distribution.

4. Association Rule Mining:

Use algorithms like **Apriori** to discover frequent itemsets and generate association rules with parameters such as support and confidence.

5. Model Evaluation:

Evaluate the performance of classification and clustering algorithms using accuracy, precision, recall, and other relevant metrics.

Visualize the results using Weka's built-in tools for decision trees, clusters, or association.

Output:-

Clustering



Classification



Association:

